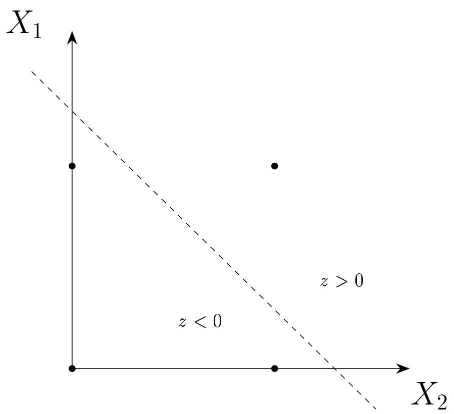
1 Introduction

Artificial intelligence is now a central part of our lives. Understanding how neural networks work is becoming increasingly important, even for individuals without a background in computer science. Considering AI systems as black boxes can lead to a passive and potentially dangerous relationship with these technologies. Large language models are complex systems that are difficult to understand. In this project, we decided to implement a simple neural network: the perceptron, using 6502 assembly language for the Commodore 64.

In machine learning, the perceptron is an algorithm used for supervised learning of binary classifiers. A binary classifier is a function that determines whether an input, represented as a vector of numbers, belongs to a specific class.[1] The perceptron implemented in this project will be trained to emulate an AND gate. To better understand the algorithm's specifications and the challenges we will encounter during implementation, we will first develop a version of the perceptron in Python. We will then translate this implementation into Commodore 64 assembly language.

We chose to implement the perceptron on the Commodore 64 for several reasons. First, we wanted to explore the architecture of a simple CPU and better understand the constraints and possibilities of low-level programming, which allows close interaction with computer hardware. The Commodore 64 was an interesting choice because of its limited instruction set, making it a suitable platform for studying the fundamentals of neural network implementation at a low level.

2 Theoretical background



x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

Figure 1: Truth table of AND gate and linear separation between the two classes.

[1] Freund, Y.; Schapire, R. E. (1999). "Large margin classification using the perceptron algorithm"